



REALISE PAR :

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Hend limème

classe

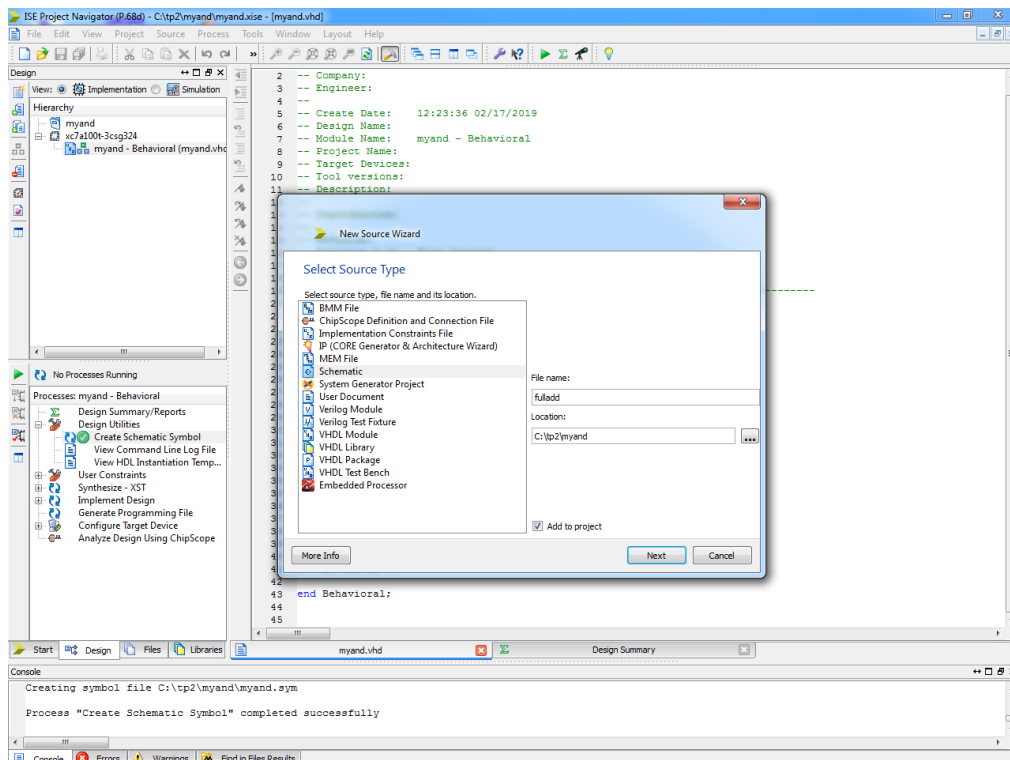
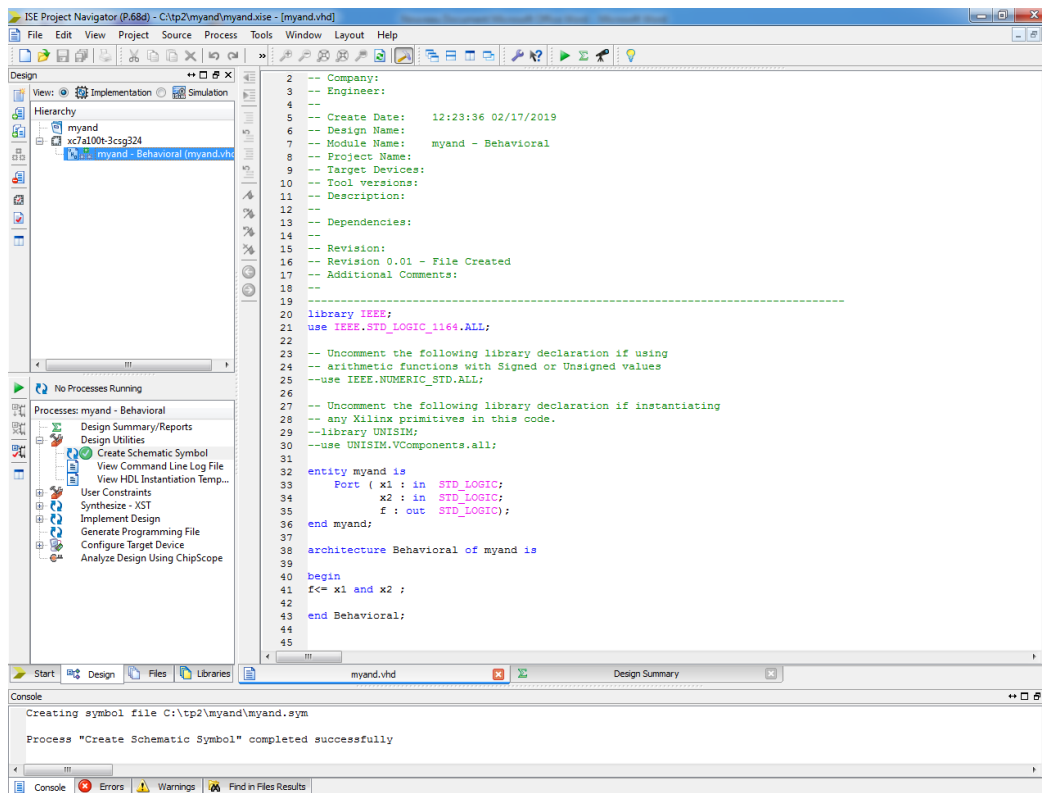
2 gec 1-1

TP2

Additionneur complet &
décodeur 1 parmi 4

- I. Création d'un projet :
- II. Création du schématique :

Code vhd du full adder :



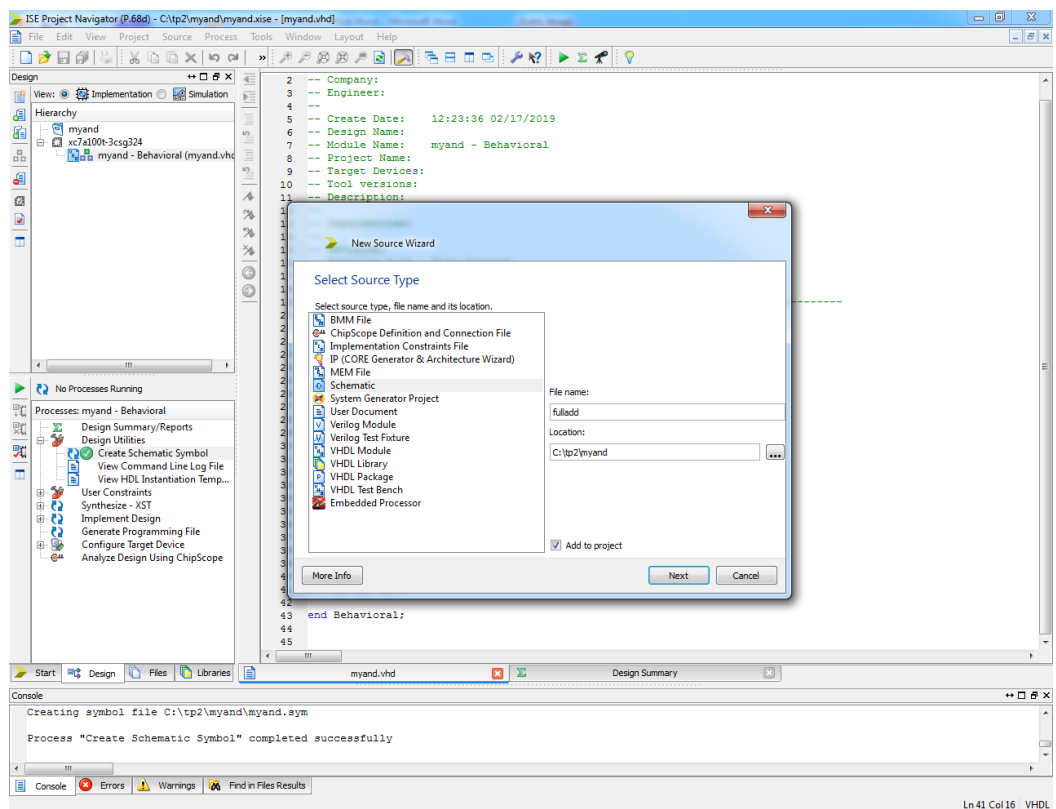
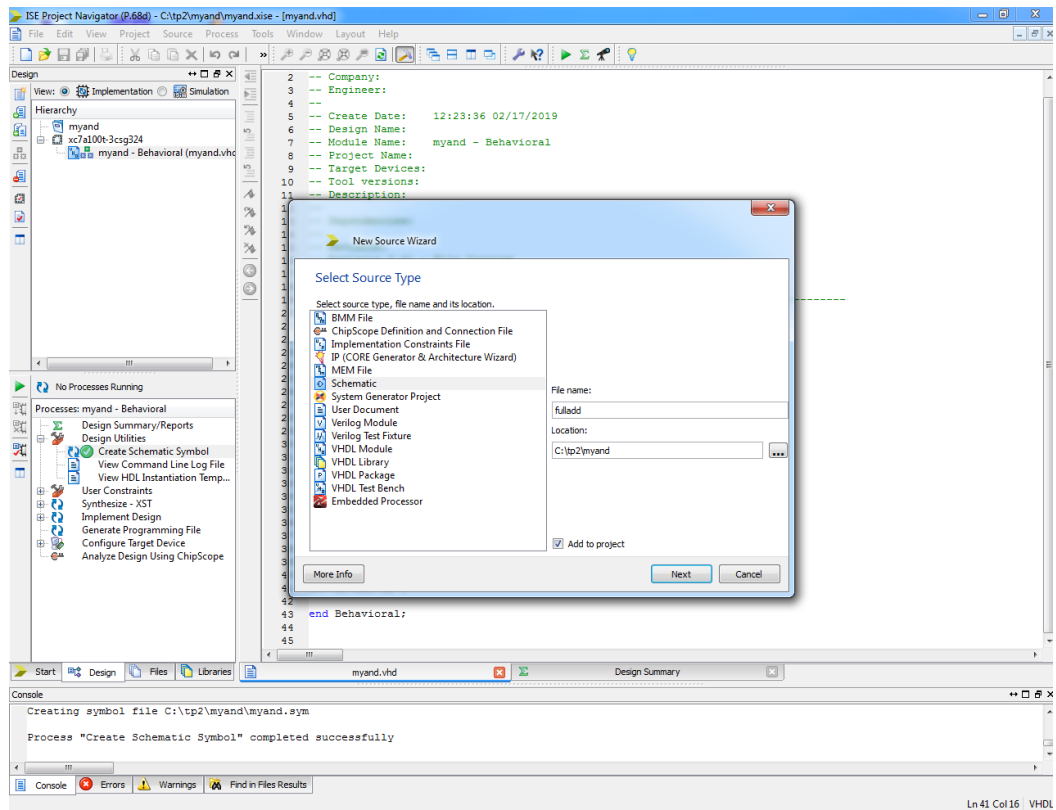
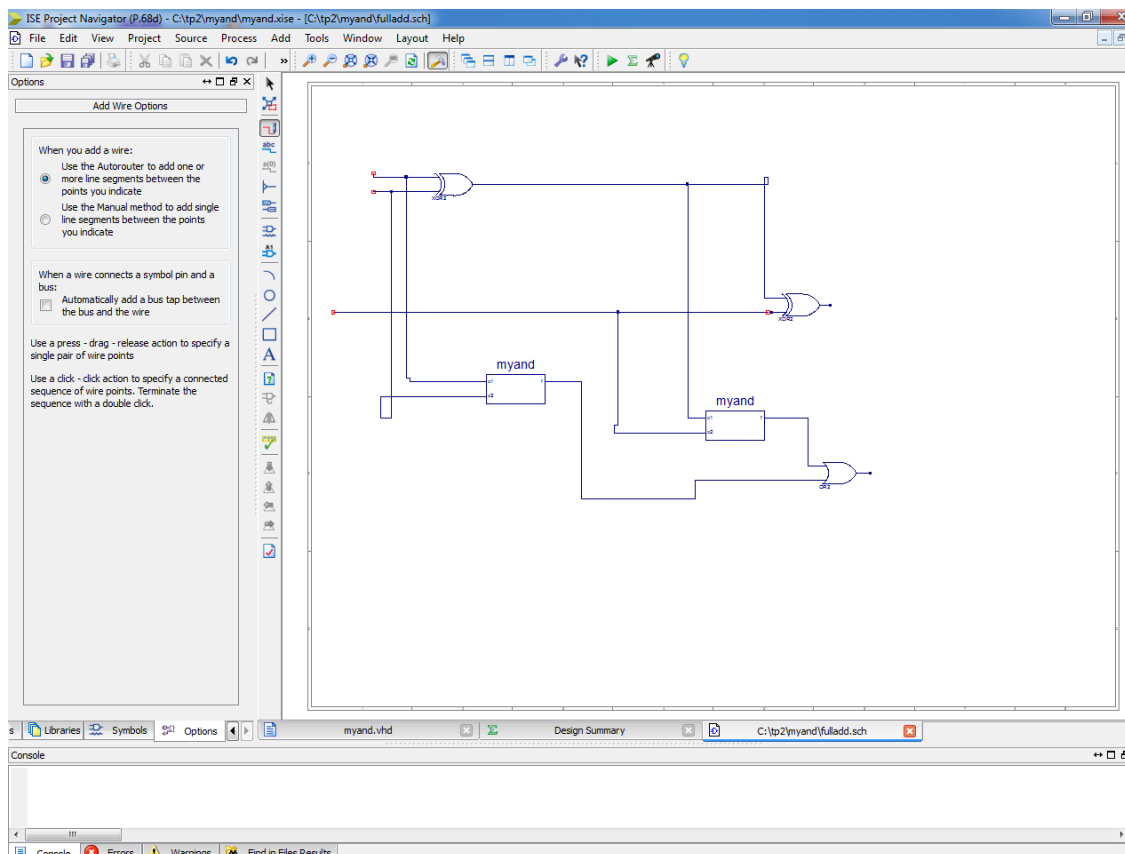
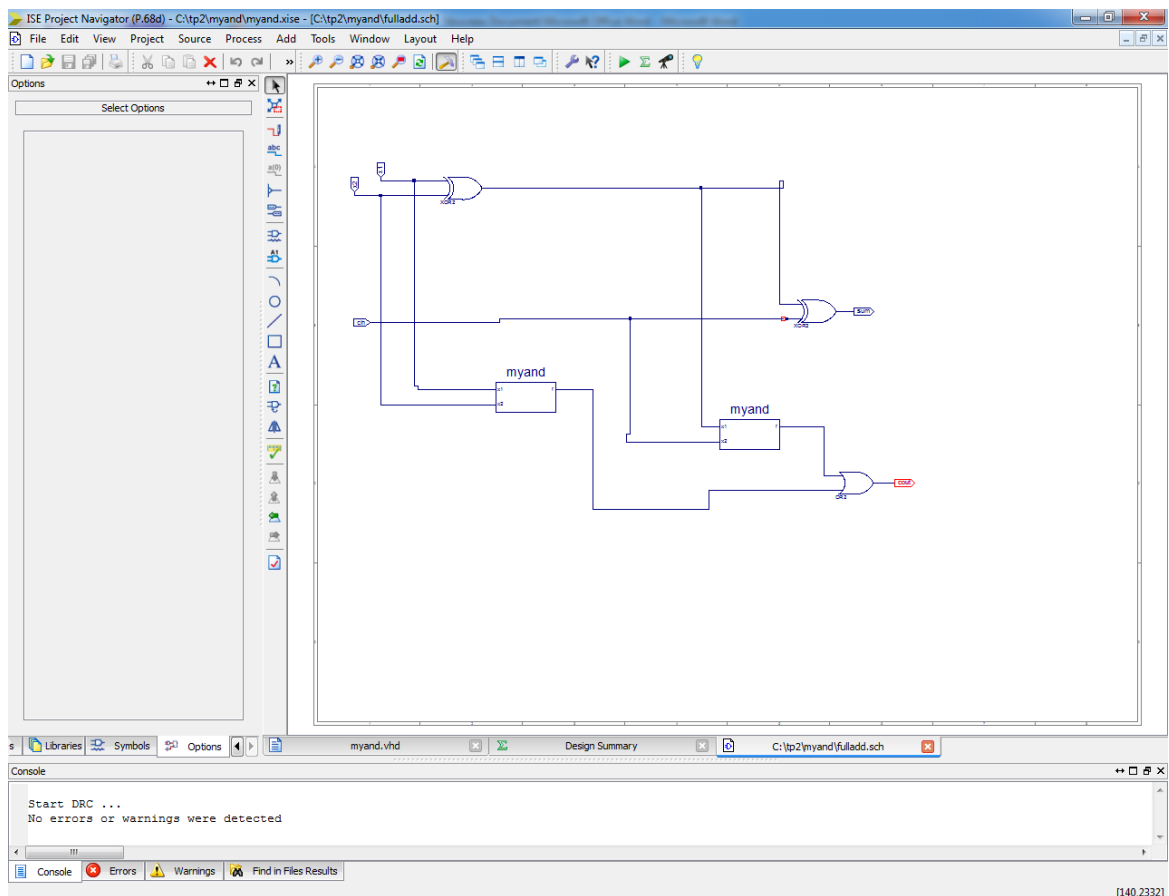


Schéma pour avoir le fonctionnement d'un ADDER 1 bit :



Ajout des entrées et des sorties



III- Simulation du circuit :

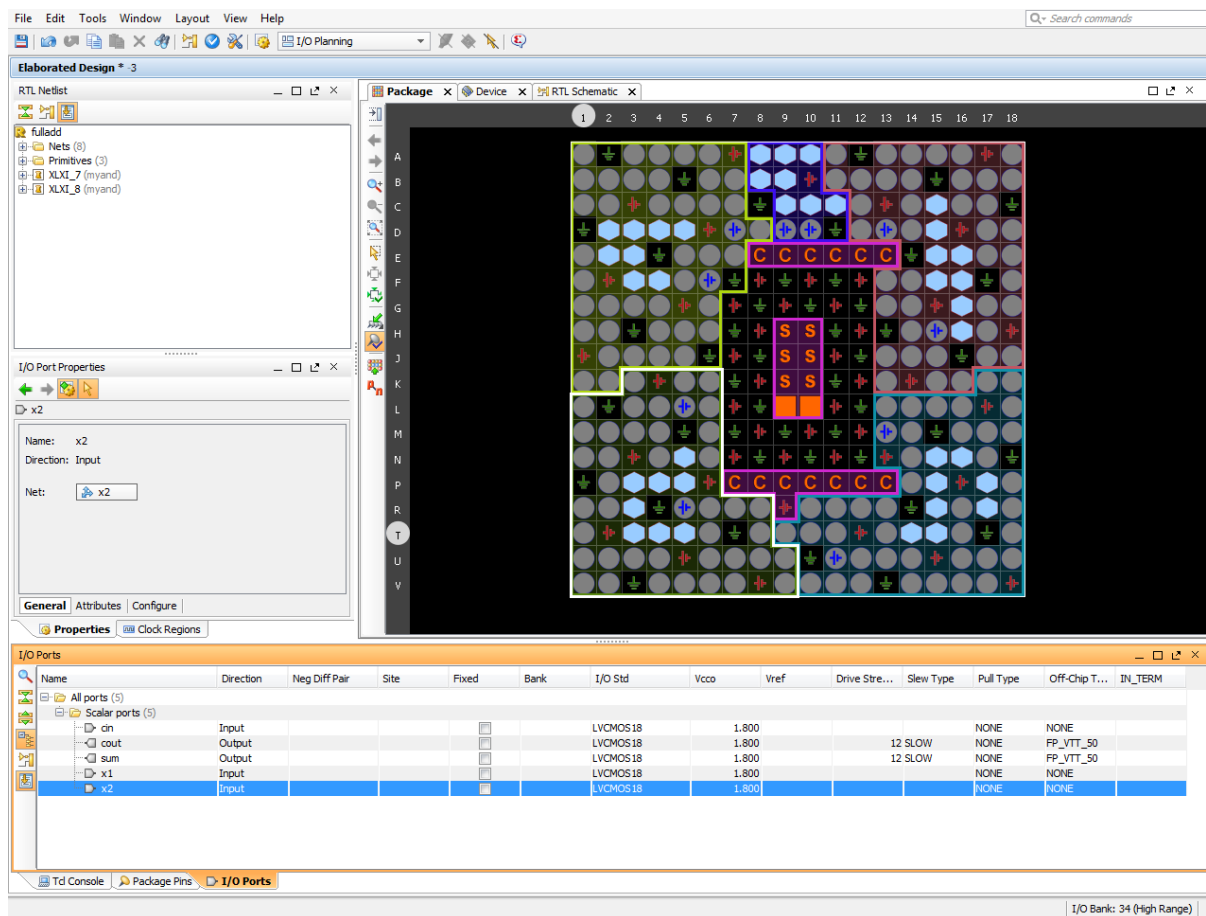
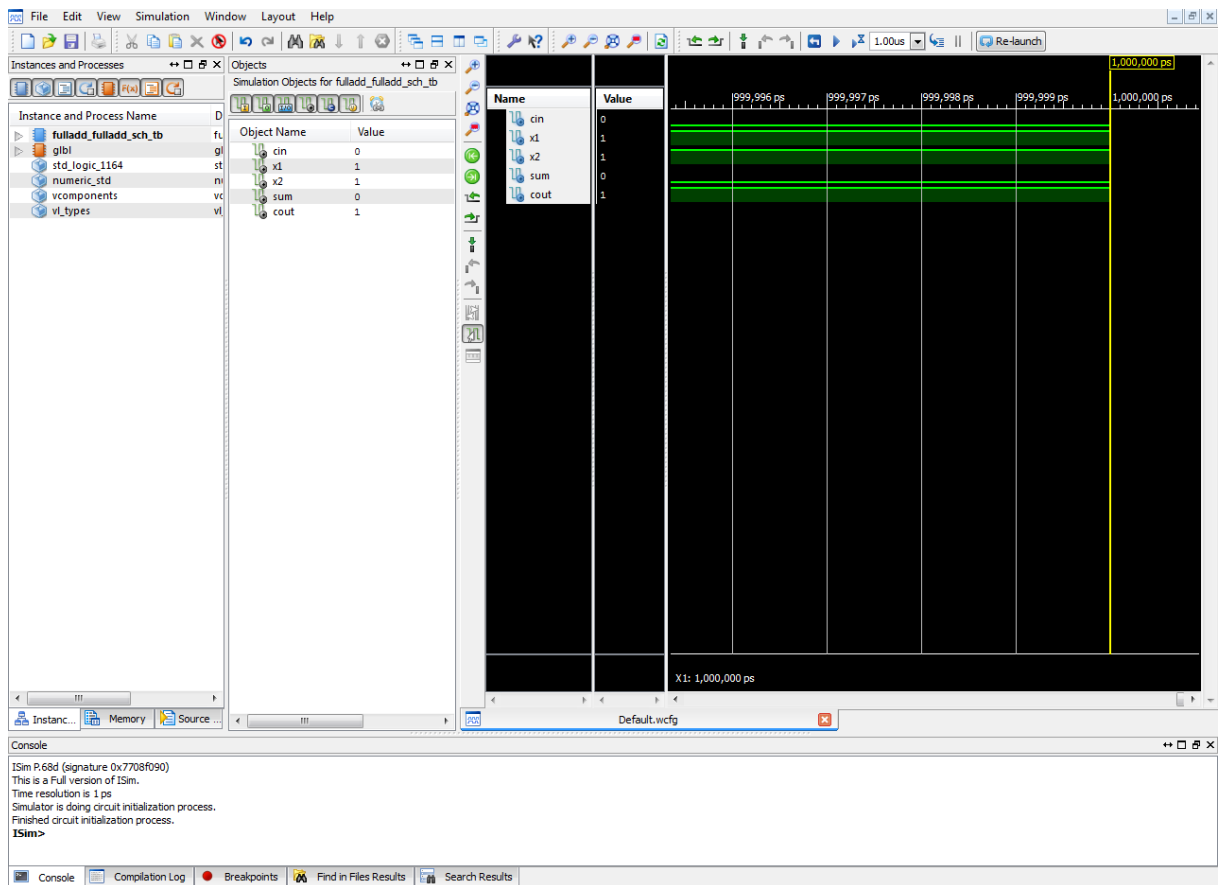
The screenshot shows the Xilinx ISE IDE interface. The top window is the 'New Source Wizard' dialog, which is used to add a new source file to the project. The 'Select Source Type' list on the left includes various file types, and 'test.vhd' is entered in the 'File name' field. The 'Location' is set to 'C:\tp2\myand'. The 'Add to project' checkbox is checked. The bottom window displays the VHDL code for the full adder and its test bench. The code is as follows:

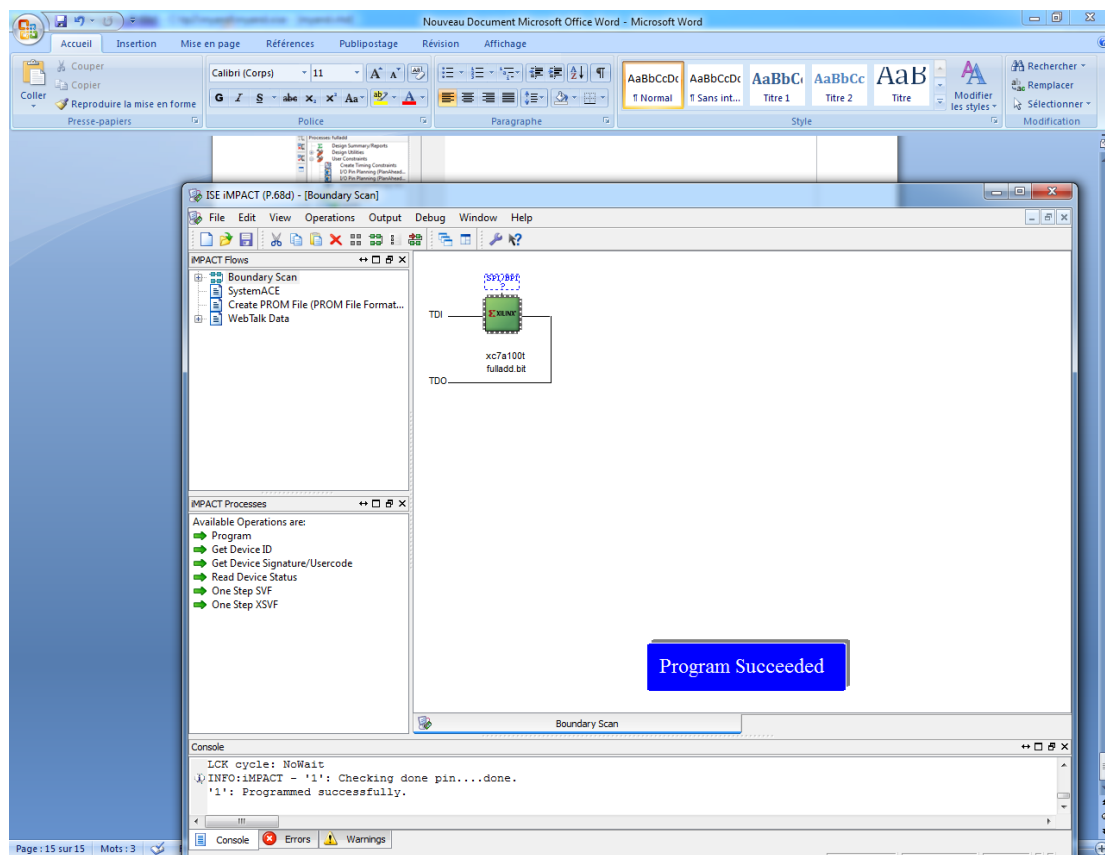
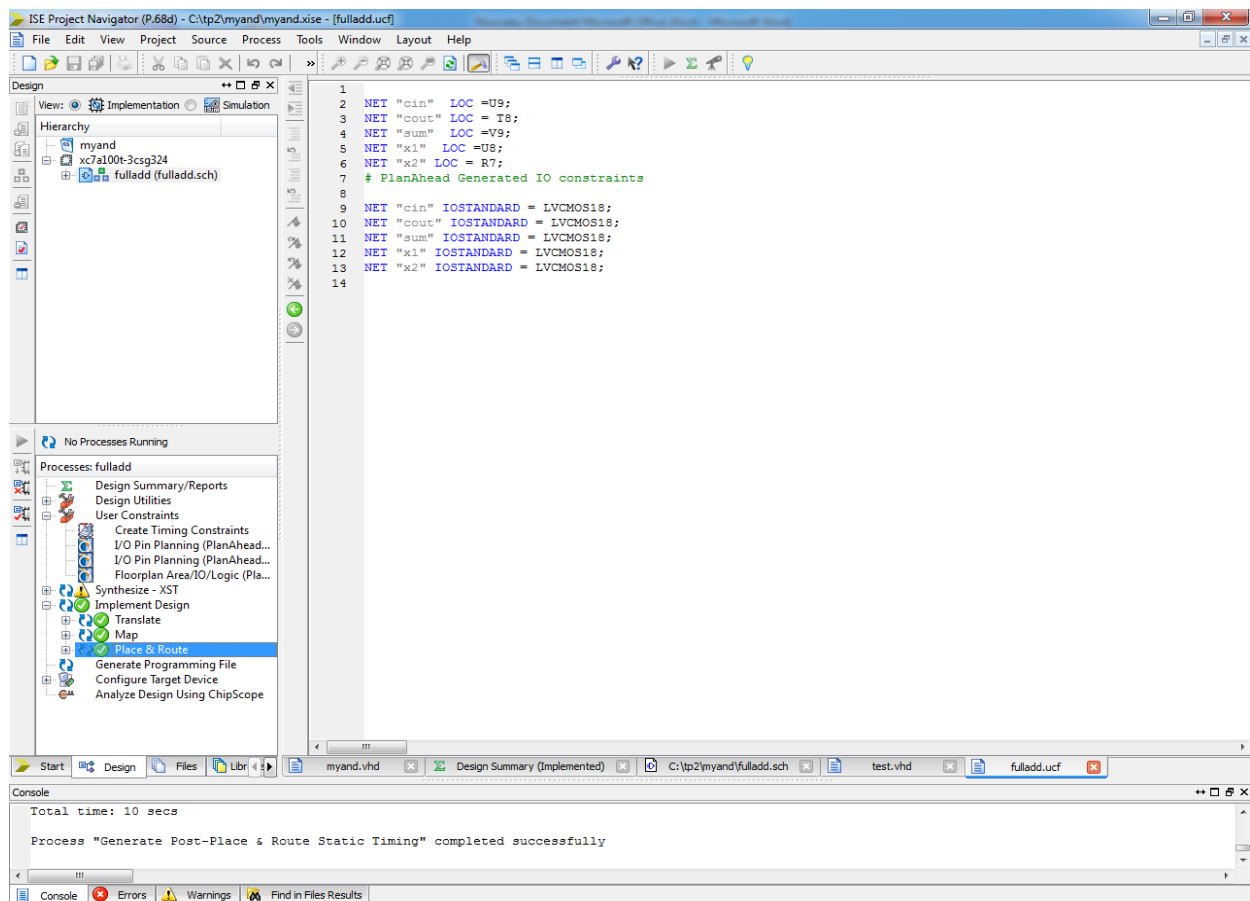
```
20 ENTITY fulladd_fulladd_sch_tb IS
21 END fulladd_fulladd_sch_tb;
22 ARCHITECTURE behavioral OF fulladd_fulladd_sch_tb IS
23
24     COMPONENT fulladd
25     PORT ( cin : IN STD_LOGIC;
26           x1  : IN STD_LOGIC;
27           x2  : IN STD_LOGIC;
28           sum : OUT STD_LOGIC;
29           cout: OUT STD_LOGIC );
30     END COMPONENT;
31
32     SIGNAL cin : STD_LOGIC;
33     SIGNAL x1  : STD_LOGIC;
34     SIGNAL x2  : STD_LOGIC;
35     SIGNAL sum : STD_LOGIC;
36     SIGNAL cout: STD_LOGIC;
37
38 BEGIN
39
40     UUT: fulladd PORT MAP (
41         cin => cin,
42         x1  => x1,
43         x2  => x2,
44         sum => sum,
45         cout => cout
46     );
47
48     -- *** Test Bench - User Defined Section ***
49     tb : PROCESS
50     BEGIN
51         x1<='1';
52         x2<='1';
53         cin<='0';
54         WAIT for 20 ns; -- will wait forever
55         x1<='0';
56         x2<='0';
57         cin<='1';
58     END PROCESS;
59     -- *** End Test Bench - User Defined Section ***
60
61 END;
```

The console window at the bottom shows the following messages:

```
Parsing VHDL file "C:/tp2/myand/test.vhd" into library isim_temp
Process "Behavioral Check Syntax" completed successfully
```

Simulation du full adder avec le test bench





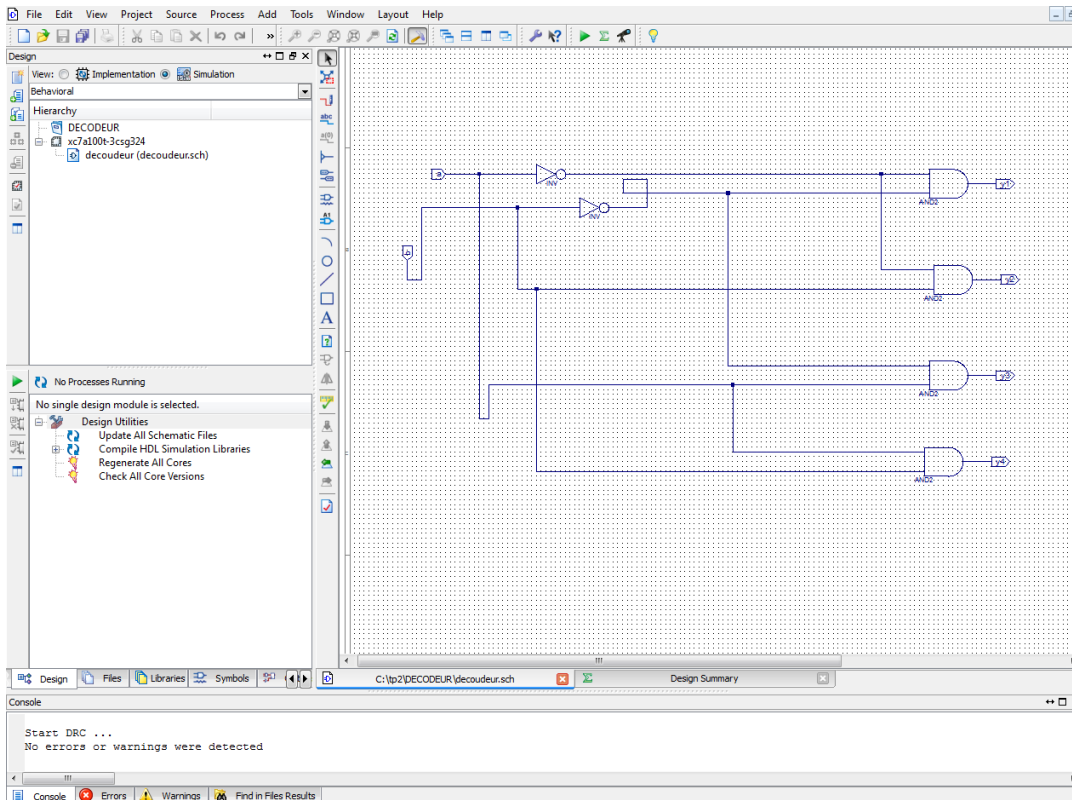
V-Travail demandé :

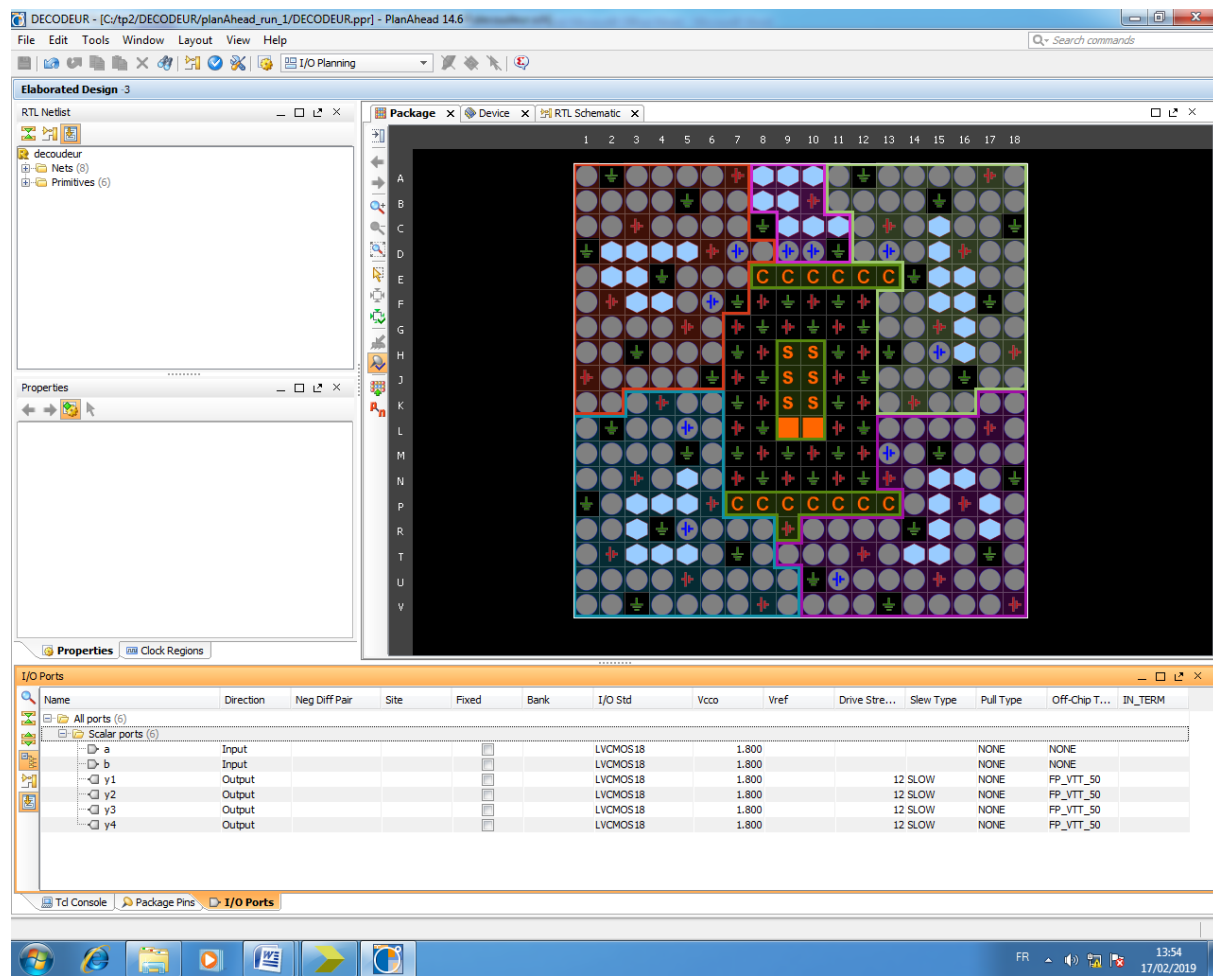
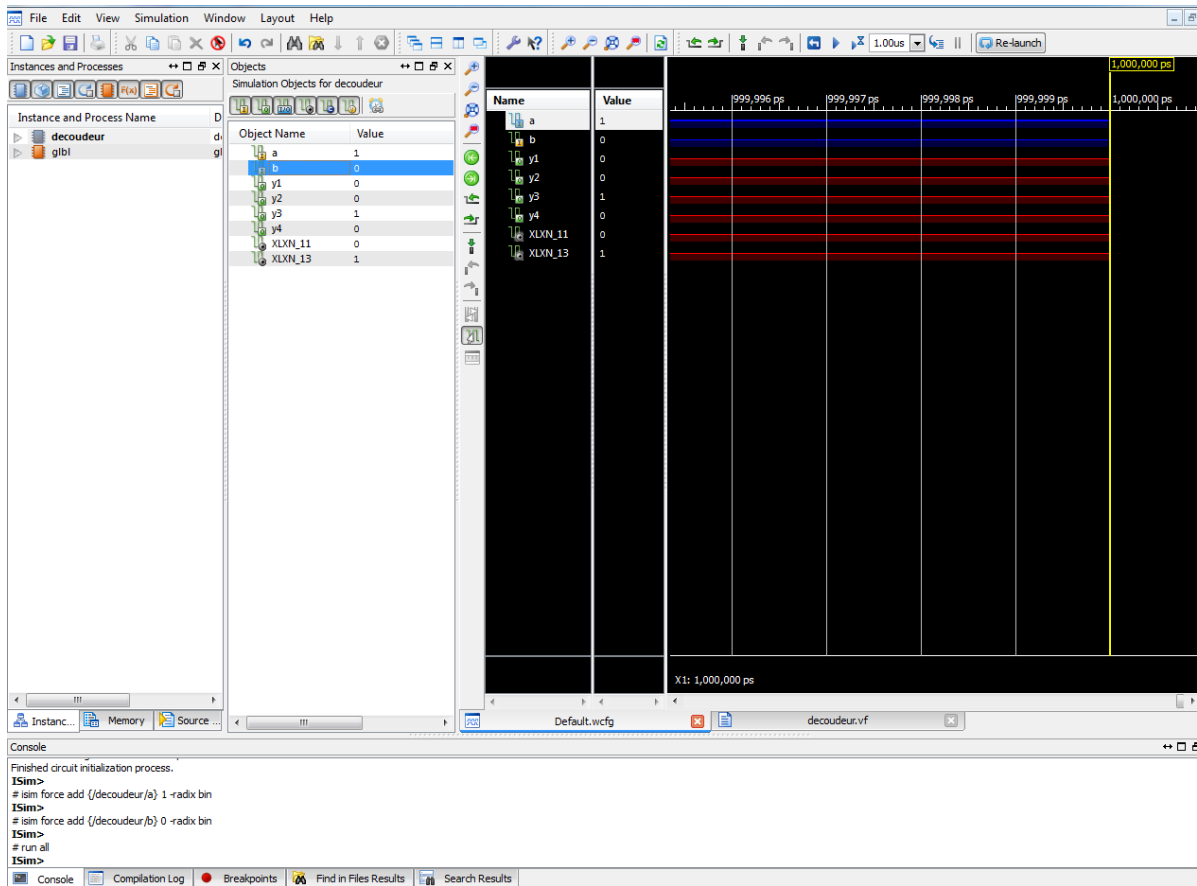
1-le fichier test bench sert à tester le fonctionnement du circuit full add

2-le fichier d'extension .ucf sert à configurer les entrées et les sorties sur la carte FPGA.

3-les outils pour programmer le FPGA sont un fichier VHDL implémenté par le logiciel Xilinx ISE

4-décodeur 1 parmi 4 :





ISE Project Navigator (P.68d) - C:\tp2\DECODEUR\DECODEUR.xise - [decodeur.ucf]

File Edit View Project Source Process Tools Window Layout Help

Design

View: Implementation Simulation

Hierarchy

- DECODEUR
 - xc7a100t-3csg324
 - decodeur (decodeur.sch)

No Processes Running

Processes: decodeur

- Create Schematic Symbol
- View Command Line Log File
- Check Design Rules
- View HDL Functional Model
- View HDL Instantiation Template
- User Constraints
 - Create Timing Constraints
 - I/O Pin Planning (PlanAhead) - Pre-Synth...
 - I/O Pin Planning (PlanAhead) - Post-Synth...
 - Floorplan Area/IO/Logic (PlanAhead)
- Synthesize - XST
- Implement Design
 - Translate
 - Map
 - Place & Route
- Generate Programming File
- Configure Target Device
- Analyze Design Using ChipScope

```
1 NET "a" LOC=U9;
2 NET "b" LOC= U8;
3 NET "y1" LOC= T8;
4 NET "y2" LOC= V9;
5 NET "y3" LOC=R8;
6 NET "y4" LOC=T6;
7 # PlanAhead Generated IO constraints
8
9
10 NET "a" IOSTANDARD = LVCMOS18;
11 NET "b" IOSTANDARD = LVCMOS18;
12 NET "y1" IOSTANDARD = LVCMOS18;
13 NET "y2" IOSTANDARD = LVCMOS18;
14 NET "y3" IOSTANDARD = LVCMOS18;
15 NET "y4" IOSTANDARD = LVCMOS18;
16
```

Console

Total time: 9 secs

Process "Generate Post-Place & Route Static Timing" completed successfully

Ln 16 Col 1 UCF

ISE Project Navigator (P.68d) - C:\tp3\afficheur\afficheur.xise - [afficheur.ucf]

File Edit View Project Source Process Tools Window Layout Help

Design

View: Implementation Simulation

Hierarchy

- afficheur
 - xc7a100t-3csg324
 - afficheur - arch (afficheur.vhd)

Running: Translate

Processes: afficheur - arch

- Design Summary/Reports
- Design Utilities
- User Constraints
 - Create Timing Constraints
 - I/O Pin Planning (PlanAhead) - Pre-Synthesis
 - I/O Pin Planning (PlanAhead) - Post-Synthesis
 - Floorplan Area/IO/Logic (PlanAhead)
- Synthesize - XST
- Implement Design
 - Translate
 - Map
 - Place & Route
- Generate Programming File
- Configure Target Device
- Analyze Design Using ChipScope

```
1
2 NET "seg[6]" LOC=L6;
3 NET "seg[5]" LOC= M2;
4 NET "seg[4]" LOC= K3;
5 NET "seg[3]" LOC=L4;
6 NET "seg[2]" LOC= L5;
7 NET "seg[1]" LOC= N1;
8 NET "seg[0]" LOC=L3;
9 NET "a[3]" LOC= R6;
10 NET "a[2]" LOC= R7;
11 NET "a[1]" LOC= U8;
12 NET "a[0]" LOC=U9;
13 # PlanAhead Generated IO constraints
14
15 NET "seg[6]" IOSTANDARD = LVCMOS18;
16 NET "seg[5]" IOSTANDARD = LVCMOS18;
17 NET "seg[4]" IOSTANDARD = LVCMOS18;
18 NET "seg[3]" IOSTANDARD = LVCMOS18;
19 NET "seg[2]" IOSTANDARD = LVCMOS18;
20 NET "seg[1]" IOSTANDARD = LVCMOS18;
21 NET "seg[0]" IOSTANDARD = LVCMOS18;
22 NET "a[3]" IOSTANDARD = LVCMOS18;
23 NET "a[2]" IOSTANDARD = LVCMOS18;
24 NET "a[1]" IOSTANDARD = LVCMOS18;
25 NET "a[0]" IOSTANDARD = LVCMOS18;
26
```

Design Summary (Synthesized)

afficheur.ucf

Console

Checking Constraint Associations...

Done...

